Yohana Bezuayene

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Education

Minnesota State University, Mankato — Mankato, MN
Bachelor of Science in Robotics Engineering | GPA: 3.9/4.0 — Expected May 2027

Relevant Coursework: Robotics Programming in C++, Dynamics, Control Systems, Embedded Systems

Technical Skills

Languages: C++, Python, Assembly, HTML5, CSS, JavaScript
Hardware & Tools: Arduino, STM32, Raspberry Pi, Sensors, Actuators, Simulink, Ubuntu, URDF, VS Code
Software: ROS2, Gazebo, MoveIt, Git, Linux, SolidWorks

Projects

ROS2 3-DOF Robot Arm Simulation — Gazebo + MoveIt — February 2026

- Designed a 3-DOF robot arm with two-fingered gripper in URDF including collision geometry and inertial properties
- Simulated full physics in Gazebo; configured MoveIt motion planning stack with custom SRDF, kinematics, and controller config
- Developed Python ROS2 node for joint state publishing enabling smooth multi-joint coordinated motion

Autonomous Line-Following Robot — PID Control — March 2026

- Built a 2WD wheeled robot using Arduino Uno, L298N dual H-bridge motor driver, and 5-channel IR sensor array
- Implemented PID controller in C++ tuning Kp, Ki, Kd parameters for stable line tracking at speed
- Designed and wired full hardware stack including motor driver, power circuit, battery management, and sensor mount

Real-Time Clock | *Embedded Systems Project* — March 2024

- Developed a real-time clock system using STM32 microcontroller, USART communication protocol, and an LCD display
- Configured hardware timers for accurate per-second display updates and implemented user time-input interface via Tera Term
- Applied I2C and SPI communication protocols for peripheral integration and hardware synchronization.

Work Experience

Undergraduate Research Assistant | *Minnesota State University, Mankato, MN* — Sep 2024 – Nov 2024

- Contributed to an autonomous robotics research project using the LEGO RCX 2.0 programmable brick.
- Developed demo programs on motor control and autonomy, and created interactive code examples to engage high school students and promote interest in STEM and robotics.

Leadership

Founder & President - IEEE Robotics & Automation Society (RAS) — Dec 2025 – Present
MNSU Student Branch Chapter | *Minnesota State University, Mankato*

- Founded the first IEEE RAS student branch chapter at Minnesota State University, Mankato
- Recruit and manage a growing membership base of robotics and engineering students
- Organize technical workshops, speaker events, and hands-on project sessions